

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AB2-041***

Frostburg 138 kV

April 2018

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Mason Dixon, LLC (“Interconnection Customer”) has proposed an incremental increase – uprate – to prior queue project AB2-038 wind generation facility located in the areas of Big Savage and Little Savage Mountains, Somerset County, Pennsylvania. The increased capability associated with this queue request AB2-041 is achieved by adding six more new wind turbines to the 19 of previous queue project AB2-038.

The following table shows the MW net contributions by Queue No. of each inverter to the Maximum Facility Output (MFO):

	Queue No.	Total # of Inverters	MW/Inverter	Total MW
	AB2-038	19	3.2	60.6
	AB2-041	6	3.2	19.4
Total MFO MW				80.0

The following table shows the MW net contributions by Queue No. of each inverter to the total Capacity Interconnection Rights (CIR) at the Point of Interconnection:

	Queue No.	Total # of Inverters	MW/Inverter	Total MW
	AB2-038	19	0.59	11.3
	AB2-041	6	0.62	3.7
Total CIR MW				15.0

AB2-038 Mason Dixon LLC Queue Request changed ownership from Big Savage LLC to Mason Dixon LLC. The original project entered the PJM queue with queue numbers U2-073/Z2-013; In the year 2017 Big Savage LLC sold its both queue requests Z2-013 and AB2-028 to Mason Dixon LLC. For more information about the history of the MFO and CIRs of this project, please refer to the AB2-038 Combined Study Report which is posted on PJM website at this link: http://www.pjm.com/pub/planning/project-queues/impact_studies/ab2038_imp.pdf.

The proposed in-service date for the AB2-041 project is December 15, 2019. **This study does not imply a Potomac Edison (“Transmission Owner”) commitment to this in-service date.**

Point of Interconnection

This project will interconnect with the Potomac Edison transmission system at same existing POI from previous projects U2-073/Z2-013 and AB2-038. This POI is a direct injection into Frostburg Substation 138 kV bus. Refer to Appendix 2 for a one-line diagram of system configuration.

Costs Summary and Transmission Owner Scope of Work

There is no additional network upgrade costs associated with this AB2-041 uprate project. All costs for attachment facilities and network upgrades necessary to support the Mason Dixon LLC AB2-038 and AB2-041 Queue projects are presented in the AB2-038 impact study report and Agreements which are published on PJM website at the following links:

Document Type	Link at PJM website
Combined Feasibility and System Impact Study Report	http://www.pjm.com/pub/planning/project-queues/impact_studies/ab2038_imp.pdf
Interconnection Service Agreement (ISA)	http://www.pjm.com/pub/planning/project-queues/impact_studies/ab2038_imp.pdf
Interconnection Construction Service Agreement (ICSA)	http://www.pjm.com/pub/planning/project-queues/csa/ab2_038_csa.pdf

Interconnection Customer Requirements

In addition to the Potomac Edison, Interconnection Customer is responsible for meeting all criteria as specified in the applicable sections of the "FirstEnergy Requirements for Transmission Connected Facilities" document, effective October 3, 2016, which can be found at this link: <http://www.pjm.com/-/media/planning/plan-standards/private-fe/fcr-facilities-connection-requirements.ashx?la=en>. Since this is an increase in output to the existing queue request AB2-038, all Interconnection Customer’s requirements stated in the Mason Dixon AB2-038 study reports and Agreements, shall apply to this AB2-041 queue project.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for Interconnection Customers's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

FirstEnergy Requirements

The Interconnection Customer will be required to comply with all FirstEnergy Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements may be found within the "FirstEnergy Requirements for Transmission Connected Facilities" document located at the following links:

<http://www.firstenergycorp.com/feconnect>

<http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>

Network Impacts

The Queue Project AB2-041 was evaluated as a 20.0 MW (Capacity 3.7 MW) injection into the Twin Ridge (Frostburg) 34.5 kV substation in the APS area. Project AB2-041 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AB2-041 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2020

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection):

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output):

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue):

None

Steady-State Voltage Requirements:

None

Short Circuit:

None

Affected System Analysis & Mitigation

NYISO Impacts:

None

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

1. (AP - AP) The 01HAZELT-01 106 J 138 kV line (from bus 235297 to bus 235305 ckt 1) loads from 104.52% to 110.95% (DC power flow) of its emergency rating (311 MVA) for the single line contingency outage of 'B2-PE-138-044_2'. This project contributes approximately 19.99 MW to the thermal violation.

CONTINGENCY 'B2-PE-138-044_2'
/* FROSTBURG - RIDGELEY HAR3 138KV
DISCONNECT BRANCH FROM BUS 917230 TO BUS 235504 CKT 1
/* 01FROSTB 138 01RIDGLY 138
END

2. (AP - AP) The 01FROSTB-Z2-038 TAP 138 kV line (from bus 235468 to bus 917230 ckt 1) loads from 102.38% to 109.08% (AC power flow) of its emergency rating (306 MVA) for the single line contingency outage of 'B2-WP-138-071'. This project contributes approximately 19.99 MW to the thermal violation.

CONTINGENCY 'B2-WP-138-071'
/* ALBRIGHT - HAZELTON - LAKE LYNN 138KV (BRANDONVILLE JUNCTION)
DISCONNECT BRANCH FROM BUS 235122 TO BUS 235305 CKT 1
/* 01LKLYNN 138 01 106 J 138
DISCONNECT BRANCH FROM BUS 235305 TO BUS 235297 CKT 1
/* 01 106 J 138 01HAZELT 138
DISCONNECT BRANCH FROM BUS 235305 TO BUS 235120 CKT 1

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/* 01 106 J 138 01ALBRIG 138
DISCONNECT BUS 235297
/* 01HAZELT 138
DISCONNECT BUS 235305
/* 01 106 J 138
END

```

3. (AP - AP) The 01GARRET-01ALBRIG 138 kV line (from bus 235469 to bus 235120 ckt 1) loads from 123.2% to 125.41% (AC power flow) of its emergency rating (191 MVA) for the single line contingency outage of 'B2-PN-115-068'. This project contributes approximately 3.23 MW to the thermal violation.

```

CONTINGENCY 'B2-PN-115-068'
/* ROCKWOOD - SOMERSET 115KV
DISCONNECT BRANCH FROM BUS 200744 TO BUS 200746 CKT 1
/* 26SOMERST 115 26ROCKWOOD 115
DISCONNECT BRANCH FROM BUS 200746 TO BUS 200773 CKT 1
/* 26ROCKWOOD 115 26ROCKWOOD 23
END

```

4. (AP - AP) The Z2-038 TAP-01RIDGLY 138 kV line (from bus 917230 to bus 235504 ckt 1) loads from 108.52% to 115.11% (AC power flow) of its emergency rating (306 MVA) for the single line contingency outage of 'B2-WP-138-071'. This project contributes approximately 19.99 MW to the thermal violation.

```

CONTINGENCY 'B2-WP-138-071'
/* ALBRIGHT - HAZELTON - LAKE LYNN 138KV (BRANDONVILLE JUNCTION)
DISCONNECT BRANCH FROM BUS 235122 TO BUS 235305 CKT 1
/* 01LKLYNN 138 01 106 J 138
DISCONNECT BRANCH FROM BUS 235305 TO BUS 235297 CKT 1
/* 01 106 J 138 01HAZELT 138
DISCONNECT BRANCH FROM BUS 235305 TO BUS 235120 CKT 1
/* 01 106 J 138 01ALBRIG 138
DISCONNECT BUS 235297
/* 01HAZELT 138
DISCONNECT BUS 235305
/* 01 106 J 138
END

```

5. (AP - AP) The AA1-047 TAP-01HAZELT 138 kV line (from bus 921081 to bus 235297 ckt 1) loads from 105.67% to 112.1% (DC power flow) of its emergency rating (311 MVA) for the single line contingency outage of 'B2-PE-138-044_2'. This project contributes approximately 19.99 MW to the thermal violation.

```

CONTINGENCY 'B2-PE-138-044_2'
/* FROSTBURG - RIDGELEY HAR3 138KV
DISCONNECT BRANCH FROM BUS 917230 TO BUS 235504 CKT 1
/* 01FROSTB 138 01RIDGLY 138
END

```

Light Load Analysis – 2020:

None.

System Reinforcements

Short Circuit:

None (No overdutied breakers identified).

Stability and Reactive Power Requirement:

None (No mitigations required).

Please refer to Appendix 3 for detailed stability study report.

Summer Peak Load Flow Analysis Reinforcements:

New System Reinforcements:

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation):

No new system reinforcements identified for this Queue Request AB2-041 (Uprate) however, all system reinforcements identified for prior Queue Request AB2-038 (Mason Dixon LLC) prevail.

Contribution to Previously Identified System Reinforcements:

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study):

None.

Light Load Load Flow Analysis Reinforcements:

New System Reinforcements:

(Upgrades required for mitigating reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation):

None.

Contribution to Previously Identified System Reinforcements:

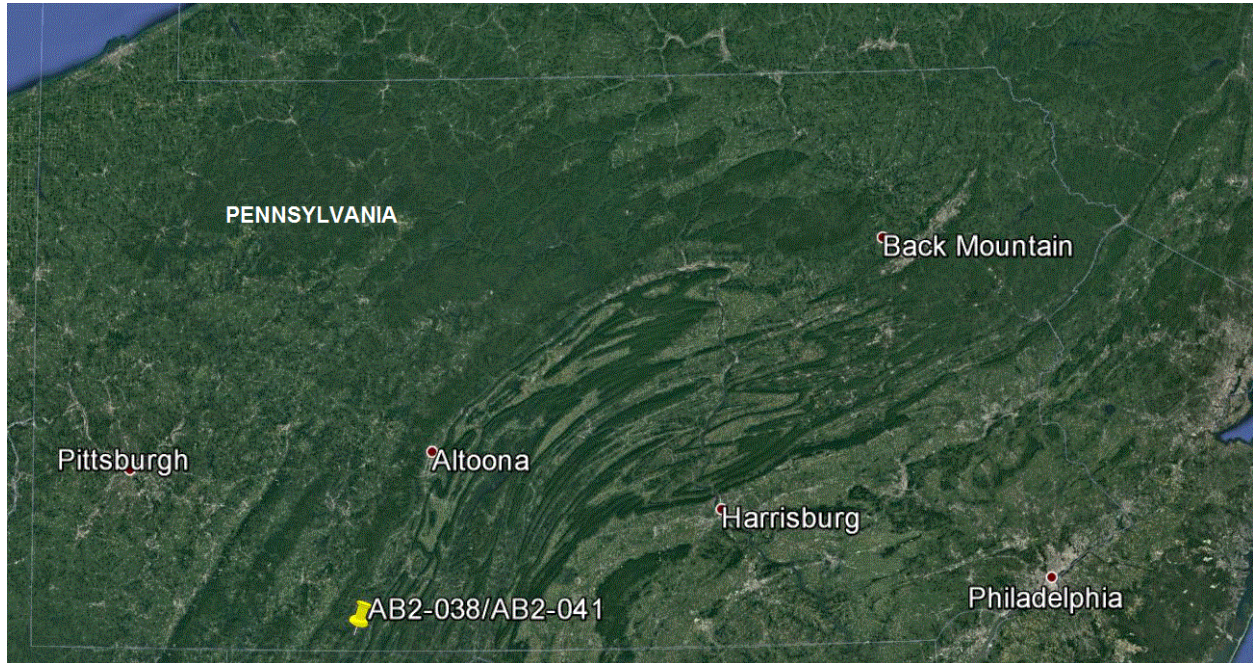
(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study):

None.

Appendix 1

Facility Location

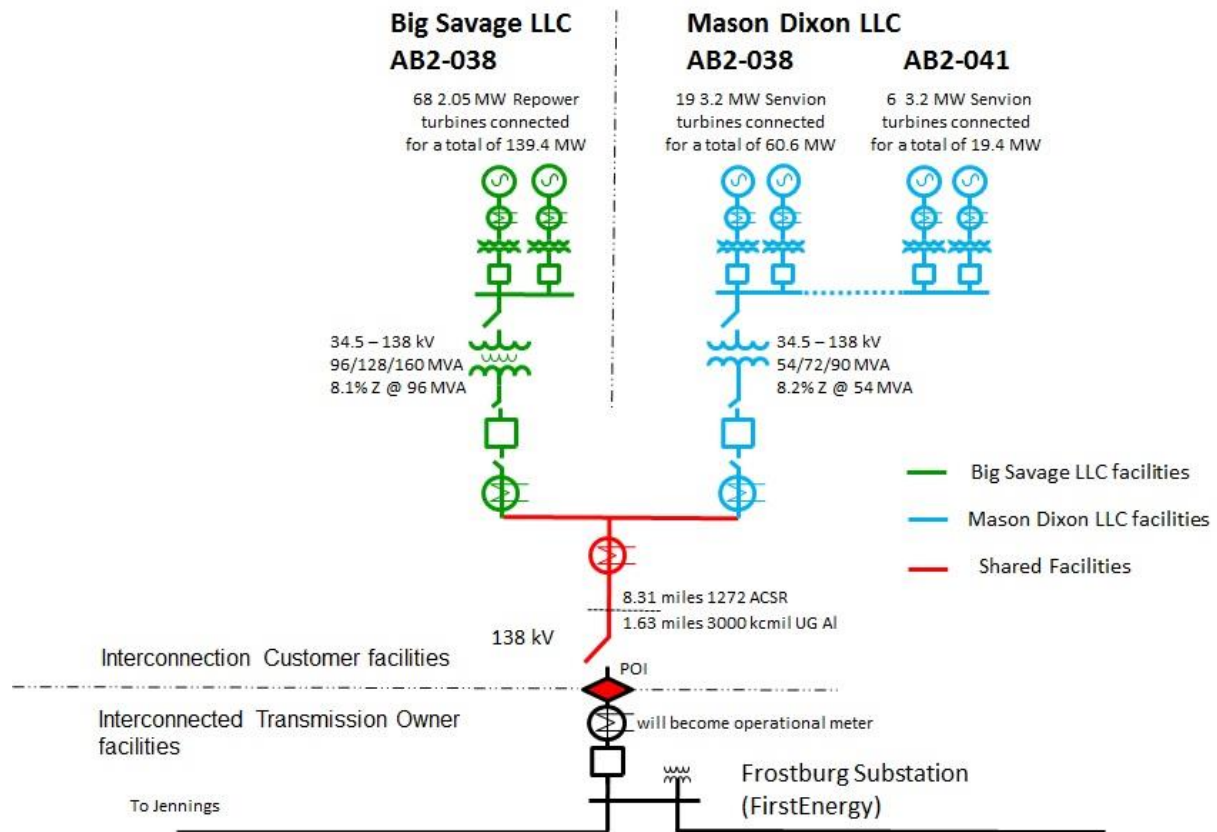
PJM Queue Position: AB2-041



Appendix 2

Interconnection One-Line Diagram

PJM Queue Position: AB2-041



Appendix 3

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

Executive Summary

Generator interconnection request U2-073/AB2-038 is for a 200 MW Maximum Facility Output (MFO) wind generating facility. U2-073/AB2-038 consists of 68 x 2.05 MW Senvion MM92 Turbines and 19 x 3.2 MW Senvion MM122 Turbines with the Point of Interconnection (POI) at the Frostburg 138 kV substation. The facility is located in the First Energy (FE) transmission system, Allegany County, Maryland.

Generator interconnection request AB2-041 is for a 19.2 MW Maximum Facility Output (MFO) wind generating facility. AB2-041 consists of 6 x 3.2 MW Senvion MM122 Turbines with the Point of Interconnection (POI) at the Frostburg 138 kV substation. The facility is located in the First Energy (FE) transmission system, Allegany County, Maryland.

This report describes a dynamic simulation analysis of U2-073/AB2-038 and AB2-041 queue projects as part of the overall system impact study.

The power flow scenario for the analysis was based on the RTEP 2020 summer peak load case, modified to include applicable queue projects. U2-073/AB2-038 and AB2-041 has been dispatched online at maximum power output, with approximately unity power factor at the high-side of the station transformer.

U2-073/AB2-038 and AB2-041 was tested for compliance with NERC, PJM, Transmission Owner and other applicable criteria. For this study, 58 contingencies were simulated, each with a 10 second simulation time period. Studied faults included:

- Steady-state operation (20 second simulation)
- Three-phase faults with normal clearing time
- Single-phase bus faults
- Single-phase faults with a stuck breaker
- Single phase faults with delayed (Zone 2) clearing at remote end due to primary relaying failure
- Single-phase faults with loss of multiple-circuit tower line

All other contingencies met the recovery criteria:

- The U2-073/AB2-038 and AB2-041 generators were able to ride through the faults except for faults where protective actions trip one or more generator(s).
- All generators maintained synchronism and any post-contingency oscillations are positively damped with a damping margin of at least 3%.
- All bus voltages recover to 0.7 p.u. within 2.5 seconds and the final voltage is within the range of 0.92 p.u. to 1.05 p.u. Note no final voltage violations less than 0.95 p.u. were observed.
- No transmission element trips, other than those either directly connected or designated to trip as a consequence of the fault.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

1. Introduction

Generator interconnection request U2-073/AB2-038 is for a 200 MW Maximum Facility Output (MFO) wind generating facility. U2-073/AB2-038 consists of 68 x 2.05 MW Senvion MM92 Turbines and 19 x 3.2 MW Senvion MM122 Turbines with the Point of Interconnection (POI) at the Frostburg 138 kV substation. The facility is located in the First Energy (FE) transmission system, Allegany County, Maryland.

Generator interconnection request AB2-041 is for a 19.2 MW Maximum Facility Output (MFO) wind generating facility. AB2-041 consists of 6 x 3.2 MW Senvion MM122 Turbines with the Point of Interconnection (POI) at the Frostburg 138 kV substation. The facility is located in the First Energy (FE) transmission system, Allegany County, Maryland.

This analysis is effectively a screening study to determine whether the addition of U2-073/AB2-038 and AB2-041 will meet the dynamic requirements of the NERC, PJM and Transmission Owner reliability standards.

This report describes the following:

- A description of the U2-073/AB2-038 and AB2-041 project and how it is proposed to be connected to the grid
- A description of how the project is modeled in this study
- A description of the fault cases analyzed in this study
- A discussion of the results.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

2. Description of the Project

U2-073 consists of 68 x 2.05 MW Senvion MM92 turbines each connected to a 34.5/0.69 kV step up transformer, all of which are connected to the 138/34.5 kV Generator Step Up transformer (GSU) of 96/128/160 MVA rating.

AB2-038 consists of 19 x 3.2 MW Senvion MM122 turbines each connected to a 34.5/0.56 kV step up transformer of 3.8 MVA rating each, all of which are connected with a 138/34.5 kV Generator Step Up transformer 54/72/90 MVA rating.

AB2-041 consists of 6 x 3.2 MW Senvion MM122 turbines each connected to a 34.5/0.56 kV step up transformer of 3.8 MVA rating each. The AB2-041 connects to the 138 kV bus through the same 138/34.5 kV GSU transformer as AB2-038.

Both generation sites are connected via same 8.31 mile generation tie line to the POI in Frostburg Substation at the 138 kV bus. Please refer to Figure 1 for a one-line diagram of the generation.

Tables 1 through 3 lists the parameters given in the impact study data and the corresponding parameters of the U2-073, AB2-038, and AB2-041 power flow models, respectively. The dynamic model for the U2-073, AB2-038, and AB2-041 plant is based on user models with the parameters supplied by the Developer.

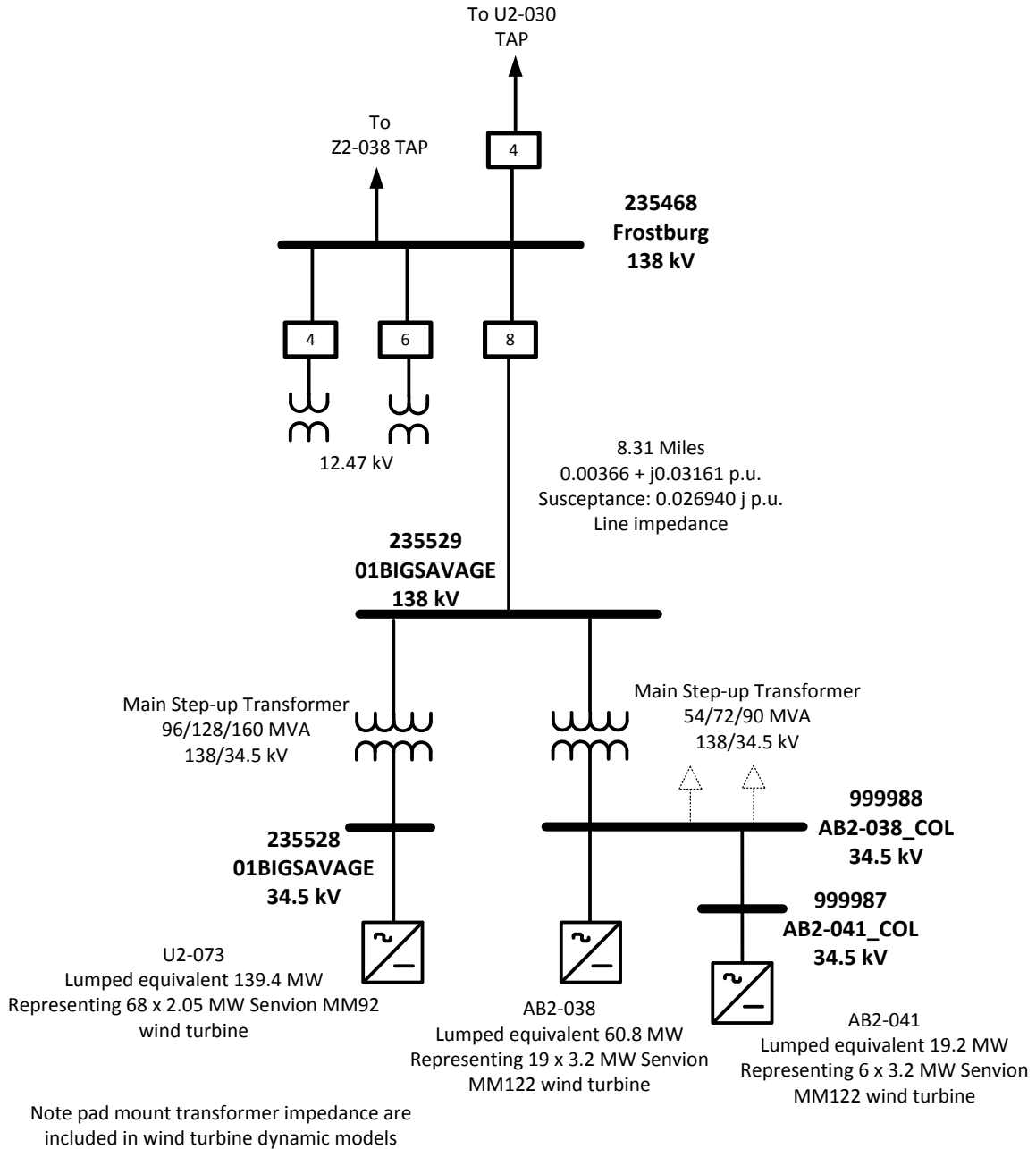


Figure 1: U2-073 and AB2-041 Plant Model

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

Table 1: U2-073 Plant Model

	Impact Study Data	Model
Wind Turbine Generators	68 x Senvion MM92 2.05 MW wind turbine generators MVA base = 2.05 MVA Vt = 0.65 kV Unsaturated sub-transient reactance = N/A	1 x Lumped equivalents representing 68 Senvion MM92 2.05 MW wind turbine generators • Pgen 139.4 MW • Pmax 139.4 MW • Pmin 0 MW • Qmax 45.819 MVar • Qmin -45.819 MVar • Mbase 139.4 MVA • Zsorce 0.0 + 0.8j pu on Mbase
Wind Turbine Transformers	58 x 34.5/0.65 kV two winding transformers Rating = Not Provided Transformer base = Not Provided Impedance = 0.00775+ j0.04651 pu Number of taps = 4 Tap step size = 2.5%	Not modeled in power flow. Generator's wind turbine dynamic model (R23A10) specified the unit transformer impedance.
Collector Transformers	1 x 138/34.5 kV three winding transformers Rating = 96/128/160 MVA Transformer base = 100 MVA Impedances = 0.004 + j0.08 pu All impedances on MVA base Number of taps = N/A Tap step size = N/A	1 x 138/34.5 kV three winding transformers Rating = 96/128/160 MVA Transformer base = 100 MVA Impedances: 0.004 + j0.08 pu All impedances on MVA base Number of taps = N/A Tap step size = N/A
Auxiliary Load	Not Provided	Not modeled
Station Load	Not Provided	Not modeled
Transmission Line	8.31 miles 138 kV line Impedance = 0.00366 + j0.03161 pu Charging susceptance = j0.02694pu on 100 MVA base	8.31 miles 138 kV line Impedance = 0.00366 + j0.03161 pu Charging susceptance = j0.02694pu on 100 MVA base

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

Table 2: AB2-038 Plant Model

	Impact Study Data	Model
Wind Turbine Generators	19 x Senvion MM122 3.2 MW wind turbine generators MVA base = 3.23 MVA Vt = 0.65 kV Unsaturated sub-transient reactance = N/A	1 x Lumped equivalents representing 19 Senvion MM122 3.2 MW wind turbine generators • Pgen 60.8 MW • Pmax 61.37 MW • Pmin 0 MW • Qmax 19.98 MVar • Qmin -19.98 MVar • Mbase 61.37 MVA • Zsorce 0.0 + 0.31j pu on Mbase
Wind Turbine Transformers	19 x 34.5/0.65 kV two winding transformers Rating = 3.8 MVA Transformer base = 3.8 MVA Impedance = 006729+ j0.062444 pu Number of taps = 5 Tap step size = 2.5%	Not modeled in power flow. Generator's wind turbine dynamic model (S33D20) specified the unit transformer impedance.
Collector Transformers	1 x 138/34.5 kV three winding transformers Rating = 54/72/90 MVA Transformer base = 54 MVA Impedances = 0.0 + j0.082 pu All impedances on MVA base Number of taps = N/A Tap step size = N/A	1 x 138/34.5 kV three winding transformers Rating = 54/72/90 MVA Transformer base = 54 MVA Impedances: 0.0 + j0.082 pu All impedances on MVA base Number of taps = N/A Tap step size = N/A
Auxiliary Load	Not Provided	Not modeled
Station Load	Not Provided	Not modeled
Transmission Line	Uses same line as U2-073.	Uses same line as U2-073.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

Table 3: AB2-041 Plant Model

	Impact Study Data	Model
Wind Turbine Generators	6 x Senvion MM122 3.2 MW wind turbine generators MVA base = 3.23 MVA Vt = 0.56 kV Unsaturated sub-transient reactance = N/A	1 x Lumped equivalents representing 6 Senvion MM122 3.2 MW wind turbine generators Pgen 19.2 MW Pmax 19.38 MW Pmin 0 MW Qmax 6.31 MVA _r Qmin -6.31 MVA _r Mbase 19.38 MVA Zsorce 0.0 + 0.31j pu on Mbase
Wind Turbine GSU Transformers	6 x 34.5/0.56 kV two winding transformers Rating = 3.8 MVA Transformer base = 3.8 MVA Impedance = 0.006729+ j0.062444 pu on MVA Base Number of taps = 5 Tap step size = 2.5%	Not modeled in power flow. Generator’s wind turbine dynamic model (S33D20) specified the unit transformer impedance.
Collector Transformers	Uses same collector transformer as AB2-038	Uses same collector transformer as AB2-038.
Auxiliary Load	Not Provided	Not modeled
Station Load	Not Provided	Not modeled
Transmission Line	Uses same line as U2-073.	Uses same line as U2-073.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

3. Loadflow and Dynamics Case Setup

The dynamic simulation analysis was performed using PSS/E Version 33.10.0.

The power flow scenario and fault cases for this study are based on PJM's Regional Transmission Planning Process¹.

The selected power flow scenario is the RTEP 2020 summer peak load case with the following modifications:

- Addition of all applicable queue projects prior to U2-073/AB2-083 and AB2-041.
- Addition of the U2-073/AB2-038 and AB2-041 queue project.
- Dispatch of units within 5 buses of the POI of the U2-073/AB2-038 and AB2-041 queue project.
 - Generators set to their maximum power output.
 - The reactive power output of each unit set near unity power factor at the high side of the station transformer.

The U2-073/AB2-038 and AB2-041 initial conditions are listed in Table , indicating maximum power output, with approximately unity power factor at the high-side of the station transformer.

Table 4: U2-073/AB2-038 and AB2-041 Machine Initial Conditions

Bus	Name	Unit	PGEN	QGEN	ETERM	POI Voltage
235528	01BIGSAVAGE	1	139.4 MW	17.84 Mvar	1.0266 pu	1.0051 pu
999988	AB2-038_COL	1	60.8 MW	15.2 Mvar	1.0439 pu	1.0051 pu
999987	AB2-041_COL	1	19.2 MW	6.3 Mvar	1.0439 pu	1.0051 pu

Generation within the vicinity of U2-073/AB2-038 and AB2-041 (5 bus radius) has been dispatched online at maximum output (P_{MAX}). The dispatch of generation in the vicinity of U2-073/AB2-038 and AB2-041 is given in Attachment 5.

¹ Manual 14B: PJM Region Transmission Planning Process, Rev 33, May 5 2016, Attachment G : PJM Stability, Short Circuit, and Special RTEP Practices and Procedures.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

4. Fault Cases

Tables 5 through 10 list the contingencies that were studied, with representative worst case total clearing times provided by PJM. Each contingency was studied over a 10 second simulation time interval.

The studied contingencies include:

- Steady-state operation (20 second simulation)
- Three-phase faults with normal clearing time
- Single-phase bus faults
- Single-phase faults with a stuck breaker
- Single phase faults with delayed (Zone 2) clearing at remote end due to primary relaying failure
- Single-phase faults with loss of multiple-circuit tower line

Contingencies were created for the following buses:

- Frostburg 138 kV
- U2-030 TAP 138 kV
- Z2-038 TAP 138 kV
- Ridgeley 138 kV
- Cumberland 138 kV
- AA1-107 138 kV

Clearing times were modeled based on PJM's "2017 Revised Clearing times for each PJM company" spreadsheet revision 20. Note FE Area 201 normal clearing times for 138 kV lines were specified as 6 or 8 to 12 cycles. All normal clearing times were initially examined as 12 cycles. If violations were observed the clearing time was reduced to 6 cycles.

Attachment 2 contains one-line diagrams of the FE network in the vicinity of U2-073/AB2-038 and AB2-041, showing where faults were applied.

The positive sequence fault impedances for single-phase faults were modeled using a zero sequence to positive sequence impedance ratio of 0.3 for FE's area derived from a separate short-circuit case. The ratio was applied to the positive sequence value in PSS/E and the single-phase fault impedance was calculated.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

5. Evaluation Criteria

This study is focused on U2-073/AB2-038 and AB2-041, along with the rest of the PJM system, maintaining synchronism and having all states return to an acceptable new condition following the disturbance. The recovery criteria applicable to this study are as per PJM's Regional Transmission Planning Process and Transmission Owner criteria:

- The U2-073/AB2-038 and AB2-041 generators were able to ride through the faults except for faults where protective actions trip one or more generator(s).
- All generators maintained synchronism and any post-contingency oscillations are positively damped with a damping margin of at least 3%.
- All bus voltages recover to 0.7 p.u. within 2.5 seconds and the final voltage is within the range of 0.92 p.u. to 1.05 p.u. Note no final voltage violations less than 0.95 p.u. were observed.
- No transmission element trips, other than those either directly connected or designated to trip as a consequence of the fault.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

6. Summary of Results

Plots from the dynamic simulations are provided in Attachment 6 with results summarized in Tables 5 through 10.

Final bus voltage violations under 0.92 p.u. were observed with that addition of the U2-073/AB2-038 and AB2-041 queue projects. Table 11 summarizes the voltage violations observed with the addition of the AB2-041 queue project.

The normal clearing times was reduced from 12 cycles to 6 cycles but did not resolve the voltage violations. Comments were provided in PJM's "2017 Revised Clearing times for each PJM company" spreadsheet revision 20 stating the typical clearing time was 6 cycles but could be as long as 8 to 12 cycles if electro-mechanical negative sequence direct ground relays are installed.

All other fault contingencies tested on the 2020 summer peak load case met the recovery criteria:

- The U2-073/AB2-038 and AB2-041 generators were able to ride through the faults except for faults where protective actions trip one or more generator(s).
- All generators maintained synchronism and any post-contingency oscillations are positively damped with a damping margin of at least 3%.
- All bus voltages recover to 0.7 p.u. within 2.5 seconds and the final voltage is within the range of 0.92 p.u. to 1.05 p.u. Note no final voltage violations less than 0.95 p.u. were observed.
- No transmission element trips, other than those either directly connected or designated to trip as a consequence of the fault.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

Table 5: Steady-State Operation

Fault Id	Fault Description	U2-073/AB2-038 and AB2-041
SS.01	Steady-State 20 second simulation.	Stable

Table 6: Three-phase Faults With Normal Clearing

Fault Id	Fault Description	Clearing Time (cycles)	U2-073/AB2-038 and AB2-041 no Mitigation
3N.01	3ph fault on Frostburg - Z2-038 Tap 138 kV line (HAR-RID)	12	Stable
3N.02	3ph fault on Frostburg - U2-030 138 kV line	12	Stable
3N.03	3ph fault on Frostburg - Big Savage 138 kV line (Trips U2-073, AB2-041, AB2-038 units)	12	Stable
3N.04	3ph fault on Z2-038 Tap - Ridgeley 138 kV line (HAR-RID)	12	Stable
3N.05	3ph fault on Ridgeley Capacitor Bank	12	Stable
3N.06	3ph fault on Ridgeley - Warrior Gen 138 kV line (Trips Warrior unit)	12	Stable
3N.07	3ph fault on Ridgeley - DansMTN 138 kV line	12	Stable
3N.08	3ph fault on Ridgeley - Cumberland 138 kV line (CU-RID)	12	Stable
3N.09	3ph fault on Ridgeley - Messick Rd 138 kV line	12	Stable
3N.10	3ph fault on Ridgeley - Hampshire 138 kV line (HS-RID)	12	Stable
3N.11	3ph fault on Ridgeley 138/34.5 KV TF 1	12	Stable
3N.12	3ph fault on Ridgeley 138/34.5 KV TF 2	12	Stable
3N.13	3ph fault on Ridgeley - Z2-038 Tap	12	Stable
3N.14	3ph fault on U2-030 - AA1-047 138 kV line (HAR-RID)	12	Stable
3N.15	3ph fault on AA1-047 - Hazelton 138 kV line (HAR-RID)	12	Stable
3N.16	3ph fault on Cumberland - Ridgeley 138 kV line (CU-RID)	12	Stable
3N.17	3ph fault on Cumberland - Black Oak 138 kV line (BO-CU)	12	Stable
3N.18	3ph fault on Cumberland 138/34.5 kV TF	12	Stable
3N.19	3ph fault on Cumberland 23.9 Mvar shunt capacitor	12	Stable

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

Table 7: Single-Phase Faults

Fault ID	Fault Description	Clearing Time (cycles)	U2-073/AB2-038 and AB2-041 no Mitigation
1S.01	SLG fault on Frostburg 138 kV bus, Loss of Frostburg 138 kV bus (Trips U2-073, AB2-038, and AB2-041 units)	12	Stable
1S.02	SLG fault on Z2-038 Tap 138 kV bus, Loss of Z2-038 Tap - Frostburg 138 kV (HAR-RID)	12	Stable
1S.03	SLG fault on Z2-038 Tap, Loss of Z2-038 Tap - Ridgeley 138 kV (HAR-RID)	12	Stable
1S.04	SLG fault on Ridgeley 138 kV bus, Loss of Ridgeley 138 kV bus	12	Stable
1S.05	SLG fault on U2-030 138 kV Bus12, Loss of U2-030 - Frostburg 138 kV	12	Stable
1S.06	SLG fault on U2-030 138 kV Bus12, Loss of U2-030 - Jennings - AA1-047 (HAR-RID)	12	Stable
1S.07	SLG fault on AA1-047 Bus, Loss of AA1-047 - Jennings - U2-030	12	Stable
1S.08	SLG fault on AA1-047 Bus, Loss of AA1-047 - Hazelton (HAR-RID)	12	Stable
1S.09	SLG fault on Cumberland 138 kV bus, Loss of Cumberland 138 kV bus	12	Stable
1S.10	Fault on Frostburg 138 kV bus. Fault cleared with loss of: - Frostburg 138 kV to U2-030 138 kV circuit 1 - Frostburg 138 kV to Ridgely 138 kV circuit 1 - Frostburg 138 kV to Big Savage 138 kV circuit 1 Contingency 'C1-PE-138-061' (Trips U2-073, AB2-038, AB2-041)	12	Stable
1S.11	Fault on Ridgely 138 kV bus. Fault cleared with loss of: - Dans Mountain 138 kV to Ridgely 138 kV circuit 1 - Cumberland 138 kV to Ridgely 138 kV circuit 1 - Z2-038 138 kV to Ridgely 138 kV circuit 1 - Messick 138 kV to Ridgely 138 kV circuit 1 - Ridgely 138 kV to Hampshire(1) 138 kV circuit 1 - Ridgely 138 kV to Ridgely CAP 138 kV circuit ZB - Ridgely 138 kV to Warrior Gen 18 kV circuit 1 - Ridgely 138 kV to Ridgely transformer 1 35 kV circuit 1 - Ridgely 138 kV to Ridgely transformer 2 35 kV circuit 2 Contingency 'C1-PE-138-021' (Trips Warrior Gen)	12	Stable

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

Table 8: Single-Phase Faults with a Stuck Breaker

Fault ID	Fault Description	Clearing Time Normal / Delayed (cycles)	U2-073/AB2-038 and AB2-041 no Mitigation
1B.01	SLG fault on Frostburg - Z2-038 TAP 138 kV, breaker 1310 stuck, Loss of Frostburg - U2-030 and Frostburg - Big Savage (Trips units U2-073, AB2-041, AB2-038)	12 / 20	Stable
1B.02	SLG fault on Frostburg - U2-030 TAP 138 kV, breaker 1310 stuck, Loss of Frostburg - Z2-038 and Frostburg - Big Savage (Trips units U2-073, AB2-041, AB2-038)	12 / 20	Stable
1B.03	SLG fault on Z2-038 Tap - Frostburg 138 kV, breaker at Z2-038 Tap, Loss of Z2-038 Tap 138 kV Bus (Trips Z2-038)	12 / 20	Stable
1B.04	SLG fault on Z2-038 Tap - Ridgeley 138 kV (HAR-RID), breaker at Z2-038 Tap stuck, Loss of Z2-038 Tap 138 kV Bus (Trips Z2-038)	12 / 20	Stable
1B.05	SLG fault on Ridgeley Capacitor Bank, breaker at Ridgeley stuck, Loss of Ridgeley 138 kV Bus	12 / 20	Stable
1B.06	SLG fault on Ridgeley - Warrior Gen, breaker at Ridgeley stuck, Loss of Ridgeley 138 kV Bus	12 / 20	Stable
1B.07	SLG fault on Ridgeley – DansMTN, breaker at Ridgeley stuck, Loss of Ridgeley 138 kV Bus	12 / 20	Stable
1B.08	SLG fault on Ridgeley - Cumberland (CU-RID), breaker at Ridgeley stuck, Loss of Ridgeley 138 kV Bus	12 / 20	Stable
1B.09	SLG fault on Ridgeley - Messick Rd , breaker at Ridgeley stuck, Loss of Ridgeley 138 kV Bus	12 / 20	Stable
1B.10	SLG fault on Ridgeley - Hampshire (HS-RID) , breaker at Ridgeley stuck, Loss of Ridgeley 138 kV Bus	12 / 20	Stable
1B.11	SLG fault on Ridgeley 138/34.5 KV TF , breaker at Ridgeley stuck, Loss of Ridgeley 138 kV Bus	12 / 20	Stable
1B.12	SLG fault on Ridgeley 138/34.5 KV TF , breaker at Ridgeley stuck, Loss of Ridgeley 138 kV Bus	12 / 20	Stable
1B.13	SLG fault on Ridgeley –Z2-038 tap , breaker at Ridgeley stuck, Loss of Ridgeley 138 kV Bus	12 / 20	Stable
1B.14	SLG fault on U2-030 - AA1-047 (HAR-RID), breaker 3 stuck. Loss of U2-030 unit (Trips U2-030 unit)	12 / 20	Stable
1B.15	SLG fault on U2-030 - Frostburg 138 kV, breaker 2 stuck. Loss of U2-030 unit (Trips U2-030 unit)	12 / 20	Stable
1B.16	SLG fault on U2-030 - Jennings - AA1-047, breaker 1 stuck. Loss of U2-030 - Frostburg, U2-030 unit. (Trips U2-030)	12 / 20	Stable
1B.17	SLG fault on AA1-047 - Hazelton (HAR-RID), breaker at AA1-047 stuck, Loss of AA1-047 unit. (Trips AA1-047)	12 / 20	Stable
1B.18	SLG fault on on Cumberland – Ridgeley 138 kV line (CU-RID), breaker RCU stuck, Loss of Cumberland 138 kV Bus	12 / 20	Stable
1B.19	SLG fault on on Cumberland – Black Oak 138 kV line (CU-BO), breaker RCU stuck, Loss of Cumberland 138 kV Bus	12 / 20	Stable
1B.20	SLG fault Cumberland 138/34.5 kV TF, breaker RCU stuck, Loss of Cumberland 138 kV Bus	12 / 20	Stable

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

Table 9: Single-Phase Faults With Delayed (Zone 2) Clearing at Remote End

Fault ID	Fault Description	Clearing Time Normal / Delayed (cycles)	U2-073/AB2-038 and AB2-041 no Mitigation
1D.01	3ph fault at 80% of 138 kV line from Frostburg - Z2-038 Tap. Delayed clearing at Frostburg.	12 / 45	Stable
1D.02	3ph fault at 80% of 138 kV line from Z2-038 Tap - Frostburg. Delayed clearing at Z2-038 Tap.	12 / 45	Stable
1D.03	3ph fault at 80% of 138 kV line from Frostburg - Big Savage. Delayed clearing at Frostburg.	12 / 45	Stable
1D.04	3ph fault at 80% of 138 kV line from Big Savage - Frostburg. Delayed clearing at Big Savage.	12 / 45	Stable
1D.05	3ph fault at 80% of 138 kV line from Frostburg - U2-030 TAP. Delayed clearing at Frostburg.	12 / 45	Stable
1D.06	3ph fault at 80% of 138 kV line from U2-030 TAP - Frostburg. Delayed clearing at U2-030 TAP.	12 / 45	Stable

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AB2-041

Table 10: Single-Phase Faults With Loss of Multiple-Circuit Tower Line

Fault ID	Fault Description	Clearing Time (cycles)	U2-073/AB2-038 and AB2-041 no Mitigation
1T.01	Fault on Ridgely 138kV circuit on Messick RD 138kV resulting in a tower failure. Fault cleared with loss of Cumberland 138kV circuit to Short Gab 138kV circuit. CONTINGENCY 'C5-PE-138-014'	12	Stable
1T.02	Fault on Ridgely 138kV circuit on Frostburg 138kV circuit resulting in tower failure. Fault cleared with loss loss of Carlos 138kV circuit. CONTINGENCY 'C5-PE-138-012_2'	12	Stable

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AB2-041 System Impact Study Report.